# Complete 332-variable quartic
# Explicit quadratic residuals

ProveIt research artifact

24 October 2026

## The single equation

Every unknown ranges over $\mathbb{N} = \{0, 1, 2, \ldots\}$, in the exact order

$$(z_0, z_1, \ldots, z_{331}).$$

The complete equation is

$$\boxed{P_{332}(z_0, \ldots, z_{331}) = \sum_{i=0}^{262} R_i(z_0, \ldots, z_{331})^2 = 0.}$$

332 variables; 263 quadratic residuals; degree 4. The collected polynomial has 11,823 monomials and coefficient height 85,243,454.

All $R_i$ are printed below, without omitted coefficients or implicit auxiliary variables. The symbols $R_i$ are polynomial abbreviations and are not additional unknowns. The index list is zero based and agrees with the supplied JSON. Original coordinate names and kinds are supplied in `variables-332.tsv`; residual names and term counts are supplied in `residuals-332.tsv`.

The report accompanying this equation proves its natural unsolvability equivalent to Con(ZFC), using the explicitly identified inherited C222 calibration. This typesetting file records the entire finite polynomial; it does not assert a new proof-assistant formalization.

Source: `equations/natural332/natural/equation.json`.
SHA-256: `e88a7abd80afaa879ae4fc5489a16684ae996bcf86d9c0c0fe82a9c54ecb7df2`.

## Complete residual list

$$R_0 = -144 - 24\,z_0 - z_0^2 + z_3$$

$$R_1 = -z_2\,z_3 + z_4$$

$$R_2 = -z_2 - z_2\,z_6 - z_2\,z_7 + z_5 + z_7$$

$$R_3 = -13 - z_0 - z_0\,z_2 - z_0\,z_9 - 13\,z_2 - 13\,z_9 + z_{11}$$

$$R_4 = -1 + z_1 - z_2 - z_2\,z_8 - 2\,z_8 - z_8\,z_9$$

$$R_5 = -1 + z_1 - z_{10} - z_{10}\,z_{11}$$

$$R_6 = 2 + z_2^2 + z_{20} - z_{21}\, z_{23}$$

$$\begin{aligned} R_7 = & -6 - 5\, z_{12} - z_{12}^2 - 2\, z_{12}\, z_{13} - 5\, z_{13} \\ & - z_{13}^2 + z_{24} \end{aligned}$$

$$\begin{aligned} R_8 = & -72 - 68\, z_{13} - 16\, z_{13}^2 - 16\, z_{13}\, z_{24} - 34\, z_{24} \\ & - 4\, z_{24}^2 + z_{25} \end{aligned}$$

$$\begin{aligned} R_9 = & -30 - 11\, z_{12} - z_{12}^2 - 4\, z_{12}\, z_{25} - 22\, z_{25} \\ & - 4\, z_{25}^2 + z_{26} \end{aligned}$$

$$\begin{aligned} R_{10} = & -380 - 312\, z_{25} - 64\, z_{25}^2 - 32\, z_{25}\, z_{26} - 78\, z_{26} \\ & - 4\, z_{26}^2 + z_{27} \end{aligned}$$

$$\begin{aligned} R_{11} = & -6 - 5\, z_{13} - z_{13}^2 - 2\, z_{13}\, z_{14} - 5\, z_{14} \\ & - z_{14}^2 + z_{28} \end{aligned}$$

$$\begin{aligned} R_{12} = & -6 - 5\, z_{12} - z_{12}^2 - 2\, z_{12}\, z_{14} - 5\, z_{14} \\ & - z_{14}^2 + z_{29} \end{aligned}$$

$$\begin{aligned} R_{13} = & -414 - 22\, z_{12} - 2\, z_{12}^2 - 16\, z_{12}\, z_{14} - 8\, z_{12}\, z_{28} \\ & - 200\, z_{13} - 32\, z_{13}^2 - 64\, z_{13}\, z_{14} - 32\, z_{13}\, z_{24} - 32\, z_{13}\, z_{29} \\ & - 320\, z_{14} - 64\, z_{14}^2 - 32\, z_{14}\, z_{24} - 32\, z_{14}\, z_{28} - 32\, z_{14}\, z_{29} \\ & - 100\, z_{24} - 8\, z_{24}^2 - 16\, z_{24}\, z_{29} - 52\, z_{28} - 8\, z_{28}^2 \\ & - 108\, z_{29} - 8\, z_{29}^2 + z_{30} \end{aligned}$$

$$\begin{aligned} R_{14} = & -1344 - 800\, z_{13} - 128\, z_{13}^2 - 256\, z_{13}\, z_{14} - 128\, z_{13}\, z_{24} \\ & - 128\, z_{13}\, z_{29} - 864\, z_{14} - 128\, z_{14}^2 - 128\, z_{14}\, z_{24} - 128\, z_{14}\, z_{29} \\ & - 400\, z_{24} - 32\, z_{24}^2 - 64\, z_{24}\, z_{29} - 432\, z_{29} - 32\, z_{29}^2 \\ & - 2\, z_{30} - 2\, z_{30}^2 + z_{31} \end{aligned}$$

$$\begin{aligned} R_{15} = & -247 - 168\, z_{12} - 32\, z_{12}^2 - 64\, z_{12}\, z_{13} - 32\, z_{12}\, z_{28} \\ & - 32\, z_{12}\, z_{29} - 188\, z_{13} - 32\, z_{13}^2 - 32\, z_{13}\, z_{28} - 32\, z_{13}\, z_{29} \\ & - 2\, z_{24} - 92\, z_{28} - 8\, z_{28}^2 - 16\, z_{28}\, z_{29} - 84\, z_{29} \\ & - 8\, z_{29}^2 + z_{32} \end{aligned}$$

$$\begin{aligned} R_{16} = & -960 - 672\, z_{12} - 128\, z_{12}^2 - 256\, z_{12}\, z_{13} - 128\, z_{12}\, z_{28} \\ & - 128\, z_{12}\, z_{29} - 736\, z_{13} - 128\, z_{13}^2 - 128\, z_{13}\, z_{28} - 128\, z_{13}\, z_{29} \\ & - 368\, z_{28} - 32\, z_{28}^2 - 64\, z_{28}\, z_{29} - 336\, z_{29} - 32\, z_{29}^2 \\ & - 2\, z_{32} - 2\, z_{32}^2 + z_{33} \end{aligned}$$

$$\begin{aligned} R_{17} = & -276 - 30\, z_{12} - 2\, z_{12}^2 - 16\, z_{12}\, z_{14} - 8\, z_{12}\, z_{28} \\ & - 30\, z_{13} - 2\, z_{13}^2 - 16\, z_{13}\, z_{14} - 8\, z_{13}\, z_{29} - 272\, z_{14} \\ & - 64\, z_{14}^2 - 32\, z_{14}\, z_{28} - 32\, z_{14}\, z_{29} - 68\, z_{28} - 8\, z_{28}^2 \\ & - 68\, z_{29} - 8\, z_{29}^2 + z_{34} \end{aligned}$$

$$
\begin{aligned}
R_{18} = & -552 - 120\,z_{13} - 8\,z_{13}^2 - 64\,z_{13}\,z_{14} - 32\,z_{13}\,z_{29} \\
& - 544\,z_{14} - 128\,z_{14}^2 - 128\,z_{14}\,z_{29} - 272\,z_{29} - 32\,z_{29}^2 \\
& - 2\,z_{34} - 2\,z_{34}^2 + z_{35}
\end{aligned}
$$

$$
\begin{aligned}
R_{19} = & -247 - 4\,z_{12} - 352\,z_{13} - 128\,z_{13}^2 - 64\,z_{13}\,z_{24} \\
& - 64\,z_{13}\,z_{28} - 92\,z_{24} - 8\,z_{24}^2 - 16\,z_{24}\,z_{28} - 84\,z_{28} \\
& - 8\,z_{28}^2 - 2\,z_{29} + z_{36}
\end{aligned}
$$

$$
\begin{aligned}
R_{20} = & -960 - 1408\,z_{13} - 512\,z_{13}^2 - 256\,z_{13}\,z_{24} - 256\,z_{13}\,z_{28} \\
& - 368\,z_{24} - 32\,z_{24}^2 - 64\,z_{24}\,z_{28} - 336\,z_{28} - 32\,z_{28}^2 \\
& - 2\,z_{36} - 2\,z_{36}^2 + z_{37}
\end{aligned}
$$

$$
\begin{aligned}
R_{21} = & -56 - 60\,z_{13} - 16\,z_{13}^2 - 16\,z_{13}\,z_{24} - 30\,z_{24} \\
& - 4\,z_{24}^2 + z_{38}
\end{aligned}
$$

$$
\begin{aligned}
R_{22} = & -30 - 11\,z_{12} - z_{12}^2 - 4\,z_{12}\,z_{38} - 22\,z_{38} \\
& - 4\,z_{38}^2 + z_{39}
\end{aligned}
$$

$$
\begin{aligned}
R_{23} = & -380 - 312\,z_{38} - 64\,z_{38}^2 - 32\,z_{38}\,z_{39} - 78\,z_{39} \\
& - 4\,z_{39}^2 + z_{40}
\end{aligned}
$$

$$
\begin{aligned}
R_{24} = & -30 - 11\,z_{13} - z_{13}^2 - 4\,z_{13}\,z_{40} - 22\,z_{40} \\
& - 4\,z_{40}^2 + z_{41}
\end{aligned}
$$

$$
\begin{aligned}
R_{25} = & -349 - 420\,z_{13} - 128\,z_{13}^2 - 128\,z_{13}\,z_{24} - 208\,z_{24} \\
& - 32\,z_{24}^2 - 2\,z_{28} + z_{42}
\end{aligned}
$$

$$
\begin{aligned}
R_{26} = & -1368 - 1664\,z_{13} - 512\,z_{13}^2 - 512\,z_{13}\,z_{24} - 832\,z_{24} \\
& - 128\,z_{24}^2 - 2\,z_{42} - 2\,z_{42}^2 + z_{43}
\end{aligned}
$$

$$
\begin{aligned}
R_{27} = & -56 - 60\,z_{12} - 16\,z_{12}^2 - 16\,z_{12}\,z_{29} - 8\,z_{12}\,z_{43} \\
& - 30\,z_{29} - 4\,z_{29}^2 - 4\,z_{29}\,z_{43} - 15\,z_{43} - z_{43}^2 \\
& + z_{44}
\end{aligned}
$$

$$
\begin{aligned}
R_{28} = & -6 - 5\,z_{14} - z_{14}^2 - 2\,z_{14}\,z_{15} - 5\,z_{15} \\
& - z_{15}^2 + z_{45}
\end{aligned}
$$

$$
\begin{aligned}
R_{29} = & -224 - 26\,z_{12} - 2\,z_{12}^2 - 16\,z_{12}\,z_{15} - 8\,z_{12}\,z_{45} \\
& - 288\,z_{14} - 128\,z_{14}^2 - 64\,z_{14}\,z_{28} - 64\,z_{14}\,z_{29} - 88\,z_{15} \\
& - 32\,z_{15}^2 - 32\,z_{15}\,z_{45} - 76\,z_{28} - 8\,z_{28}^2 - 16\,z_{28}\,z_{29} \\
& - 68\,z_{29} - 8\,z_{29}^2 - 44\,z_{45} - 8\,z_{45}^2 + z_{46}
\end{aligned}
$$

$$
\begin{aligned}
R_{30} = & -640 - 1152\,z_{14} - 512\,z_{14}^2 - 256\,z_{14}\,z_{28} - 256\,z_{14}\,z_{29} \\
& - 304\,z_{28} - 32\,z_{28}^2 - 64\,z_{28}\,z_{29} - 272\,z_{29} - 32\,z_{29}^2 \\
& - 2\,z_{46} - 2\,z_{46}^2 + z_{47}
\end{aligned}
$$

$$
\begin{aligned}
R_{31} = & -241 - 168\,z_{12} - 32\,z_{12}^2 - 64\,z_{12}\,z_{13} - 32\,z_{12}\,z_{28} \\
& - 32\,z_{12}\,z_{29} - 184\,z_{13} - 32\,z_{13}^2 - 32\,z_{13}\,z_{28} - 32\,z_{13}\,z_{29} \\
& - z_{14} - 92\,z_{28} - 8\,z_{28}^2 - 16\,z_{28}\,z_{29} - 84\,z_{29} \\
& - 8\,z_{29}^2 + z_{48}
\end{aligned}
$$

$$
\begin{aligned}
R_{32} = & -42 - 52\,z_{14} - 16\,z_{14}^2 - 16\,z_{14}\,z_{28} - 26\,z_{28} \\
& - 4\,z_{28}^2 + z_{49}
\end{aligned}
$$

$$
\begin{aligned}
R_{33} = & -30 - 11\,z_{14} - z_{14}^2 - 4\,z_{14}\,z_{49} - 22\,z_{49} \\
& - 4\,z_{49}^2 + z_{50}
\end{aligned}
$$

$$
\begin{aligned}
R_{34} = & -306 - 35\,z_{13} - z_{13}^2 - 16\,z_{13}\,z_{49} - 4\,z_{13}\,z_{50} \\
& - 280\,z_{49} - 64\,z_{49}^2 - 32\,z_{49}\,z_{50} - 70\,z_{50} - 4\,z_{50}^2 \\
& + z_{51}
\end{aligned}
$$

$$
\begin{aligned}
R_{35} = & -4556 - 4320\,z_{49} - 1024\,z_{49}^2 - 512\,z_{49}\,z_{50} - 128\,z_{49}\,z_{51} \\
& - 1080\,z_{50} - 64\,z_{50}^2 - 32\,z_{50}\,z_{51} - 270\,z_{51} - 4\,z_{51}^2 \\
& + z_{52}
\end{aligned}
$$

$$
\begin{aligned}
R_{36} = & -30 - 11\,z_{14} - z_{14}^2 - 4\,z_{14}\,z_{52} - 22\,z_{52} \\
& - 4\,z_{52}^2 + z_{53}
\end{aligned}
$$

$$
\begin{aligned}
R_{37} = & -306 - 35\,z_{13} - z_{13}^2 - 16\,z_{13}\,z_{52} - 4\,z_{13}\,z_{53} \\
& - 280\,z_{52} - 64\,z_{52}^2 - 32\,z_{52}\,z_{53} - 70\,z_{53} - 4\,z_{53}^2 \\
& + z_{54}
\end{aligned}
$$

$$
\begin{aligned}
R_{38} = & -1028 - 672\,z_{12} - 128\,z_{12}^2 - 256\,z_{12}\,z_{13} - 128\,z_{12}\,z_{28} \\
& - 128\,z_{12}\,z_{29} - 736\,z_{13} - 128\,z_{13}^2 - 128\,z_{13}\,z_{28} - 128\,z_{13}\,z_{29} \\
& - 368\,z_{28} - 32\,z_{28}^2 - 64\,z_{28}\,z_{29} - 336\,z_{29} - 32\,z_{29}^2 \\
& - 2\,z_{48} - 2\,z_{48}^2 - 32\,z_{52} - 8\,z_{53} - 2\,z_{54} \\
& + z_{55}
\end{aligned}
$$

$$
\begin{aligned}
R_{39} = & -264 - 128\,z_{52} - 32\,z_{53} - 8\,z_{54} - 2\,z_{55} \\
& - 2\,z_{55}^2 + z_{56}
\end{aligned}
$$

$$
\begin{aligned}
R_{40} = & -2 - 3\,z_{12} - z_{12}^2 - 2\,z_{12}\,z_{56} - 3\,z_{56} \\
& - z_{56}^2 + z_{57}
\end{aligned}
$$

$$
\begin{aligned}
R_{41} = & -12 - 28\,z_{56} - 16\,z_{56}^2 - 16\,z_{56}\,z_{57} - 14\,z_{57} \\
& - 4\,z_{57}^2 + z_{58}
\end{aligned}
$$

$$
\begin{aligned}
R_{42} = & -30 - 11\,z_{14} - z_{14}^2 - 4\,z_{14}\,z_{58} - 22\,z_{58} \\
& - 4\,z_{58}^2 + z_{59}
\end{aligned}
$$

$$
\begin{aligned}
R_{43} = & -306 - 35\,z_{12} - z_{12}^2 - 16\,z_{12}\,z_{58} - 4\,z_{12}\,z_{59} \\
& - 280\,z_{58} - 64\,z_{58}^2 - 32\,z_{58}\,z_{59} - 70\,z_{59} - 4\,z_{59}^2 \\
& + z_{60}
\end{aligned}
$$

$$
\begin{aligned}
R_{44} = & -4556 - 4320\,z_{58} - 1024\,z_{58}^2 - 512\,z_{58}\,z_{59} - 128\,z_{58}\,z_{60} \\
& - 1080\,z_{59} - 64\,z_{59}^2 - 32\,z_{59}\,z_{60} - 270\,z_{60} - 4\,z_{60}^2 \\
& + z_{61}
\end{aligned}
$$

$$
\begin{aligned}
R_{45} = & -90 - 76\,z_{14} - 16\,z_{14}^2 - 16\,z_{14}\,z_{28} - 16\,z_{14}\,z_{61} \\
& - 38\,z_{28} - 4\,z_{28}^2 - 8\,z_{28}\,z_{61} - 38\,z_{61} - 4\,z_{61}^2 \\
& + z_{62}
\end{aligned}
$$

$$
\begin{aligned}
R_{46} = & -4556 - 4320\,z_{52} - 1024\,z_{52}^2 - 512\,z_{52}\,z_{53} - 128\,z_{52}\,z_{54} \\
& - 1080\,z_{53} - 64\,z_{53}^2 - 32\,z_{53}\,z_{54} - 270\,z_{54} - 4\,z_{54}^2 \\
& + z_{63}
\end{aligned}
$$

$$
\begin{aligned}
R_{47} = & -682 - 70\,z_{13} - 2\,z_{13}^2 - 32\,z_{13}\,z_{61} - 8\,z_{13}\,z_{62} \\
& - 592\,z_{61} - 128\,z_{61}^2 - 64\,z_{61}\,z_{62} - 148\,z_{62} - 8\,z_{62}^2 \\
& - 2\,z_{63} + z_{64}
\end{aligned}
$$

$$
R_{48} = -z_{64} - z_{64}^2 + z_{65}
$$

$$
\begin{aligned}
R_{49} = & -42 - 52\,z_{13} - 16\,z_{13}^2 - 8\,z_{13}\,z_{14} - 16\,z_{13}\,z_{24} \\
& - 13\,z_{14} - z_{14}^2 - 4\,z_{14}\,z_{24} - 26\,z_{24} - 4\,z_{24}^2 \\
& + z_{66}
\end{aligned}
$$

$$
\begin{aligned}
R_{50} = & -2278 - 2264\,z_{13} - 800\,z_{13}^2 - 800\,z_{13}\,z_{24} - 160\,z_{13}\,z_{66} \\
& - 70\,z_{14} - 2\,z_{14}^2 - 32\,z_{14}\,z_{63} - 8\,z_{14}\,z_{65} - 1132\,z_{24} \\
& - 200\,z_{24}^2 - 80\,z_{24}\,z_{66} - 592\,z_{63} - 128\,z_{63}^2 - 64\,z_{63}\,z_{65} \\
& - 148\,z_{65} - 8\,z_{65}^2 - 228\,z_{66} - 8\,z_{66}^2 + z_{67}
\end{aligned}
$$

$$
\begin{aligned}
R_{51} = & -2712 - 280\,z_{14} - 8\,z_{14}^2 - 128\,z_{14}\,z_{63} - 32\,z_{14}\,z_{65} \\
& - 2368\,z_{63} - 512\,z_{63}^2 - 256\,z_{63}\,z_{65} - 592\,z_{65} - 32\,z_{65}^2 \\
& - 2\,z_{67} - 2\,z_{67}^2 + z_{68}
\end{aligned}
$$

$$
R_{52} = -2 - 3\,z_{68} - z_{68}^2 + z_{69}
$$

$$
R_{53} = -z_2 - z_2\,z_{17} + z_{70}
$$

$$
R_{54} = -z_{16}^2 + 2\,z_{16}\,z_{17} - z_{17}^2 + z_{71}
$$

$$
R_{55} = -2\,z_{27}\,z_{70} - 4\,z_{70} + z_{72}
$$

$$
\begin{aligned}
R_{56} = & -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{72} - z_{17}^2 - 2\,z_{17}\,z_{72} \\
& - z_{72}^2 + z_{73}
\end{aligned}
$$

$$
R_{57} = -z_{31}\,z_{70} + z_{74}
$$

$$
\begin{aligned}
R_{58} = & -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{74} - z_{17}^2 - 2\,z_{17}\,z_{74} \\
& - z_{74}^2 + z_{75}
\end{aligned}
$$

$$R_{59} = -z_{33}\,z_{70} + z_{76}$$

$$R_{60} = -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{76} - z_{17}^2 - 2\,z_{17}\,z_{76} - z_{76}^2 + z_{77}$$

$$R_{61} = -z_{35}\,z_{70} + z_{78}$$

$$R_{62} = -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{78} - z_{17}^2 - 2\,z_{17}\,z_{78} - z_{78}^2 + z_{79}$$

$$R_{63} = -z_{37}\,z_{70} + z_{80}$$

$$R_{64} = -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{80} - z_{17}^2 - 2\,z_{17}\,z_{80} - z_{80}^2 + z_{81}$$

$$R_{65} = -4\,z_{13}\,z_{70} - 2\,z_{41}\,z_{70} - 4\,z_{70} + z_{82}$$

$$R_{66} = -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{82} - z_{17}^2 - 2\,z_{17}\,z_{82} - z_{82}^2 + z_{83}$$

$$R_{67} = -4\,z_{43}\,z_{70} - 2\,z_{44}\,z_{70} + z_{84}$$

$$R_{68} = -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{84} - z_{17}^2 - 2\,z_{17}\,z_{84} - z_{84}^2 + z_{85}$$

$$R_{69} = -z_{47}\,z_{70} + z_{86}$$

$$R_{70} = -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{86} - z_{17}^2 - 2\,z_{17}\,z_{86} - z_{86}^2 + z_{87}$$

$$R_{71} = -z_{12}^2 + 2\,z_{12}\,z_{14} - z_{14}^2 + z_{88}$$

$$R_{72} = -z_{13}^2 + 2\,z_{13}\,z_{14} - z_{14}^2 + z_{89}$$

$$R_{73} = -z_{88}\,z_{89} + z_{90}$$

$$R_{74} = -4 - 4\,z_{15} - z_{15}^2 + z_{91}$$

$$R_{75} = -z_{90}^2 + 2\,z_{90}\,z_{91} - z_{91}^2 + z_{92}$$

$$R_{76} = -2\,z_{69}\,z_{70} - 4\,z_{70} + z_{93}$$

$$R_{77} = -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{93} - z_{17}^2 - 2\,z_{17}\,z_{93} - z_{93}^2 + z_{94}$$

$$R_{78} = -z_2 - z_2\,z_{13} + z_{95}$$

$$R_{79} = -1 - z_{17} - z_{17}\,z_{95} - z_{95} + z_{96}$$

$$R_{80} = -z_2 - z_2\,z_{12} + z_{97}$$

$$R_{81} = -4\,z_2 - 4\,z_2\,z_{13} - 2\,z_2\,z_{24} + z_{98}$$

$$R_{82} = -1 - z_{16} - z_{16}\,z_{97} - z_{97} + z_{99}$$

$$R_{83} = -z_{98}\,z_{99} - z_{99} + z_{100}$$

$$R_{84} = -z_{96}^2 + 2\,z_{96}\,z_{100} - z_{100}^2 + z_{101}$$

$$R_{85} = -6\,z_2 - 4\,z_2\,z_{13} - 2\,z_2\,z_{24} + z_{102}$$

$$R_{86} = -1 - z_{17} - z_{17}\,z_{102} - z_{102} + z_{103}$$

$$R_{87} = -1 - z_{16} - z_{16}\,z_{95} - z_{95} + z_{104}$$

$$R_{88} = -z_{103}^2 + 2\,z_{103}\,z_{104} - z_{104}^2 + z_{105}$$

$$R_{89} = -z_{71}\,z_{73} + z_{106}$$

$$R_{90} = -z_{75}\,z_{106} + z_{107}$$

$$R_{91} = -z_{77}\,z_{107} + z_{108}$$

$$R_{92} = -z_{79}\,z_{108} + z_{109}$$

$$R_{93} = -z_{81}\,z_{109} + z_{110}$$

$$R_{94} = -z_{83}\,z_{110} + z_{111}$$

$$R_{95} = -z_{85}\,z_{111} + z_{112}$$

$$R_{96} = -z_{87}\,z_{112} + z_{113}$$

$$R_{97} = -z_{92}\,z_{113} - z_{94}\,z_{113} + z_{114}$$

$$R_{98} = -z_{101}\,z_{114} + z_{115}$$

$$R_{99} = -2 - 2\,z_2 - z_2\,z_{20} - 2\,z_9 - z_9\,z_{20} - z_{20} + z_{116}$$

$$R_{100} = -1 - z_{18} - z_{18}\,z_{116} - z_{116} + z_{117}$$

$$
\begin{aligned}
R_{101} = & -2 - z_2 - z_2\,z_{19} - z_9 - z_9\,z_{19} \\
& - 2\,z_{19} - z_{19}\,z_{116} - z_{116} + z_{118}
\end{aligned}
$$

$$
\begin{aligned}
R_{102} = & \, 2 - 4\,z_1 + 2\,z_1^2 - 2\,z_1\,z_{16} - 2\,z_1\,z_{17} \\
& - 2\,z_1\,z_{117} - 2\,z_1\,z_{118} + 2\,z_{16} + z_{16}^2 + 2\,z_{16}\,z_{117} \\
& + 2\,z_{17} + z_{17}^2 + 2\,z_{17}\,z_{118} - z_{22}\,z_{119} + z_{105}\,z_{115} \\
& + 2\,z_{117} + z_{117}^2 + 2\,z_{118} + z_{118}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{103} = & -9 - z_{12} - z_{13} - z_{14} - z_{15} \\
& - z_{16} - z_{17} - z_{18} - z_{19} - z_{20} \\
& + 9\,z_{22}^2 - z_{120}
\end{aligned}
$$

$$
R_{104} = 2 + z_6 + z_7 - z_{22}\,z_{121} + z_{188}
$$

$$
\begin{aligned}
R_{105} = & -1 - z_6 - z_6\,z_{186} - z_6\,z_{187} - z_7 \\
& - z_7\,z_{186} - z_7\,z_{187} - 2\,z_{186} - z_{186}\,z_{188} - 2\,z_{187} \\
& - z_{187}\,z_{188} - z_{188} + z_{211} + z_{212}
\end{aligned}
$$

$$
R_{106} = -z_{22} - z_{22}\,z_{279} - z_{22}\,z_{280} + z_{122}
$$

$$
\begin{aligned}
R_{107} = & -54 - 3\,z_0 - 3\,z_0\,z_{125} - 3\,z_1 - 3\,z_1\,z_{125} \\
& - 5\,z_2 - 5\,z_2\,z_{125} - z_4 - z_4\,z_{125} - 2\,z_5 \\
& - 2\,z_5\,z_{125} - z_6 - z_6\,z_{125} - z_7 - z_7\,z_{125} \\
& - 3\,z_9 - 3\,z_9\,z_{125} - z_{21} - z_{21}\,z_{125} - 2\,z_{22} \\
& - 2\,z_{22}\,z_{125} - z_{121} - z_{121}\,z_{125} - 54\,z_{125} - z_{125}\,z_{186} \\
& - z_{125}\,z_{187} - z_{125}\,z_{188} - z_{125}\,z_{256} - z_{125}\,z_{257} - z_{125}\,z_{279} \\
& - z_{125}\,z_{280} + z_{126} - z_{186} - z_{187} - z_{188} \\
& - z_{256} - z_{257} - z_{279} - z_{280}
\end{aligned}
$$

$$
\begin{aligned}
R_{108} = & -3\,z_0\,z_{126} - 3\,z_1\,z_{126} - 5\,z_2\,z_{126} - z_4\,z_{126} - 2\,z_5\,z_{126} \\
& - z_6\,z_{126} - z_7\,z_{126} - 3\,z_9\,z_{126} - z_{21}\,z_{126} - 2\,z_{22}\,z_{126} \\
& - z_{121}\,z_{126} - 55\,z_{126} - z_{126}\,z_{186} - z_{126}\,z_{187} - z_{126}\,z_{188} \\
& - z_{126}\,z_{256} - z_{126}\,z_{257} - z_{126}\,z_{279} - z_{126}\,z_{280} + z_{127}
\end{aligned}
$$

$$
R_{109} = -1 + z_{123}^2 + z_{126}^2 - z_{127}^2
$$

$$
\begin{aligned}
R_{110} = & -10 - 6\,z_6 - z_6^2 - 2\,z_6\,z_7 + 2\,z_6\,z_{123} \\
& + 2\,z_6\,z_{124} - 2\,z_6\,z_{188} - 6\,z_7 - z_7^2 + 2\,z_7\,z_{123} \\
& + 2\,z_7\,z_{124} - 2\,z_7\,z_{188} - z_{122}\,z_{128} + 6\,z_{123} + 2\,z_{123}\,z_{188} \\
& + 6\,z_{124} + 2\,z_{124}\,z_{188} - 6\,z_{188} - z_{188}^2
\end{aligned}
$$

$$
R_{111} = -z_{123}\,z_{130} - z_{124}\,z_{130} + z_{133}
$$

$$
R_{112} = -1 + z_{129}^2 + z_{130}^2 - z_{133}^2
$$

$$
R_{113} = -4\,z_{130} - 4\,z_{130}\,z_{131} + z_{134}
$$

$$R_{114} = 1 - z_{123} - z_{124} - z_{129}\, z_{132} + z_{134}$$

$$\begin{aligned} R_{115} = & - z_0\, z_{123} - z_0\, z_{124} - 12\, z_{123} - z_{123}\, z_{138} - 12\, z_{124} \\ & - z_{124}\, z_{138} + z_{139} \end{aligned}$$

$$\begin{aligned} R_{116} = 143 & + 24\, z_0 + z_0^2 + 2\, z_0\, z_{138} + z_{135}^2 \\ & + 24\, z_{138} + z_{138}^2 - z_{139}^2 \end{aligned}$$

$$R_{117} = - z_{134}\, z_{137} - z_{137} + z_{140}$$

$$R_{118} = - 1 + z_{136}^2 + z_{137}^2 - z_{140}^2$$

$$R_{119} = - z_{129}\, z_{141} - z_{135} + z_{136}$$

$$\begin{aligned} R_{120} = & - 12 - z_0 - 4\, z_0\, z_{142} + z_{137} - 4\, z_{138}\, z_{142} \\ & - 48\, z_{142} \end{aligned}$$

$$\begin{aligned} R_{121} = & - 2 - 2\, z_4 - z_4^2 + 2\, z_4\, z_{123} + 2\, z_4\, z_{124} \\ & + 2\, z_{123} + 2\, z_{124} - z_{143} \end{aligned}$$

$$\begin{aligned} R_{122} = 12 & + z_0 + z_0\, z_4 + 12\, z_4 + z_4\, z_{138} \\ & - z_5 + z_{135} + z_{138} - z_{139} - z_{143}\, z_{144} \end{aligned}$$

$$\begin{aligned} R_{123} = & - 2\, z_0\, z_{123} - 2\, z_0\, z_{124} - 2\, z_1\, z_{123} - 2\, z_1\, z_{124} - 4\, z_2\, z_{123} \\ & - 4\, z_2\, z_{124} - 2\, z_9\, z_{123} - 2\, z_9\, z_{124} - 30\, z_{123} - z_{123}\, z_{148} \\ & - 30\, z_{124} - z_{124}\, z_{148} + z_{149} \end{aligned}$$

$$\begin{aligned} R_{124} = 899 & + 120\, z_0 + 4\, z_0^2 + 8\, z_0\, z_1 + 16\, z_0\, z_2 \\ & + 8\, z_0\, z_9 + 4\, z_0\, z_{148} + 120\, z_1 + 4\, z_1^2 + 16\, z_1\, z_2 \\ & + 8\, z_1\, z_9 + 4\, z_1\, z_{148} + 240\, z_2 + 16\, z_2^2 + 16\, z_2\, z_9 \\ & + 8\, z_2\, z_{148} + 120\, z_9 + 4\, z_9^2 + 4\, z_9\, z_{148} + z_{145}^2 \\ & + 60\, z_{148} + z_{148}^2 - z_{149}^2 \end{aligned}$$

$$R_{125} = - z_{134}\, z_{147} - z_{147} + z_{150}$$

$$R_{126} = - 1 + z_{146}^2 + z_{147}^2 - z_{150}^2$$

$$R_{127} = - z_{129}\, z_{151} - z_{145} + z_{146}$$

$$\begin{aligned} R_{128} = & - 30 - 2\, z_0 - 8\, z_0\, z_{152} - 2\, z_1 - 8\, z_1\, z_{152} \\ & - 4\, z_2 - 16\, z_2\, z_{152} - 2\, z_9 - 8\, z_9\, z_{152} + z_{147} \\ & - 4\, z_{148}\, z_{152} - 120\, z_{152} \end{aligned}$$

$$R_{129} = - 1 - z_{21}^2 + 2\, z_{21}\, z_{123} + 2\, z_{21}\, z_{124} - z_{153}$$

$$\begin{aligned} R_{130} = & \, 2\, z_0\, z_{21} + 2\, z_1\, z_{21} + 4\, z_2\, z_{21} + 2\, z_9\, z_{21} + 30\, z_{21} \\ & + z_{21}\, z_{148} - z_{22} + z_{145} - z_{149} - z_{153}\, z_{154} \end{aligned}$$

$$
\begin{aligned}
R_{131} = & -2 - 2\,z_5 - z_5^2 - 2\,z_5\,z_{22} + 2\,z_5\,z_{123} \\
& + 2\,z_5\,z_{124} - 2\,z_{22} - z_{22}^2 + 2\,z_{22}\,z_{123} + 2\,z_{22}\,z_{124} \\
& + 2\,z_{123} + 2\,z_{124} - z_{155}
\end{aligned}
$$

$$
\begin{aligned}
R_{132} = & \; 30 + 2\,z_0 + 2\,z_0\,z_5 + 2\,z_0\,z_{22} + 2\,z_1 \\
& + 2\,z_1\,z_5 + 2\,z_1\,z_{22} + 4\,z_2 + 4\,z_2\,z_5 + 4\,z_2\,z_{22} \\
& + 30\,z_5 + 2\,z_5\,z_9 + z_5\,z_{148} + 2\,z_9 + 2\,z_9\,z_{22} \\
& + 30\,z_{22} + z_{22}\,z_{148} - z_{121} + z_{145} + z_{148} \\
& - z_{149} - z_{155}\,z_{156}
\end{aligned}
$$

$$
\begin{aligned}
R_{133} = & -5 - 4\,z_6 - z_6^2 - 2\,z_6\,z_7 + 2\,z_6\,z_{123} \\
& + 2\,z_6\,z_{124} - 2\,z_6\,z_{188} - 4\,z_7 - z_7^2 + 2\,z_7\,z_{123} \\
& + 2\,z_7\,z_{124} - 2\,z_7\,z_{188} + 4\,z_{123} + 2\,z_{123}\,z_{188} + 4\,z_{124} \\
& + 2\,z_{124}\,z_{188} - z_{157} - 4\,z_{188} - z_{188}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{134} = & \; 23 + 2\,z_0 + z_0\,z_6 + z_0\,z_7 + z_0\,z_{188} \\
& + 12\,z_6 + z_6\,z_{138} + 12\,z_7 + z_7\,z_{138} + z_{135} \\
& + 2\,z_{138} + z_{138}\,z_{188} - z_{139} - z_{157}\,z_{158} - z_{186} \\
& - z_{187} + 12\,z_{188}
\end{aligned}
$$

$$
\begin{aligned}
R_{135} = & -z_2\,z_{123} - z_2\,z_{124} - z_9\,z_{123} - z_9\,z_{124} - 2\,z_{123} \\
& - z_{123}\,z_{162} - 2\,z_{124} - z_{124}\,z_{162} + z_{163}
\end{aligned}
$$

$$
\begin{aligned}
R_{136} = & \; 3 + 4\,z_2 + z_2^2 + 2\,z_2\,z_9 + 2\,z_2\,z_{162} \\
& + 4\,z_9 + z_9^2 + 2\,z_9\,z_{162} + z_{159}^2 + 4\,z_{162} \\
& + z_{162}^2 - z_{163}^2
\end{aligned}
$$

$$
R_{137} = -z_{134}\,z_{161} - z_{161} + z_{164}
$$

$$
R_{138} = -1 + z_{160}^2 + z_{161}^2 - z_{164}^2
$$

$$
R_{139} = -z_{129}\,z_{165} - z_{159} + z_{160}
$$

$$
\begin{aligned}
R_{140} = & -2 - z_2 - 4\,z_2\,z_{166} - z_9 - 4\,z_9\,z_{166} \\
& + z_{161} - 4\,z_{162}\,z_{166} - 8\,z_{166}
\end{aligned}
$$

$$
\begin{aligned}
R_{141} = & \; 3 + 2\,z_2 + z_2\,z_6 + z_2\,z_7 + z_2\,z_{188} \\
& + 2\,z_6 + z_6\,z_9 + z_6\,z_{162} + 2\,z_7 + z_7\,z_9 \\
& + z_7\,z_{162} + 2\,z_9 + z_9\,z_{188} - z_{157}\,z_{167} + z_{159} \\
& + 2\,z_{162} + z_{162}\,z_{188} - z_{163} + 2\,z_{188} - z_{256} \\
& - z_{257}
\end{aligned}
$$

$$
\begin{aligned}
R_{142} = & -z_1\,z_{123} - z_1\,z_{124} - z_{123} - z_{123}\,z_{171} - z_{124} \\
& - z_{124}\,z_{171} + z_{172}
\end{aligned}
$$

$$
\begin{aligned}
R_{143} = & \; 2\,z_1 + z_1^2 + 2\,z_1\,z_{171} + z_{168}^2 + 2\,z_{171} \\
& + z_{171}^2 - z_{172}^2
\end{aligned}
$$

$$R_{144} = -z_{134}\,z_{170} - z_{170} + z_{173}$$

$$R_{145} = -1 + z_{169}^2 + z_{170}^2 - z_{173}^2$$

$$R_{146} = -z_{129}\,z_{174} - z_{168} + z_{169}$$

$$\begin{aligned} R_{147} = {} & -1 - z_1 - 4\,z_1\,z_{175} + z_{170} - 4\,z_{171}\,z_{175} \\ & - 4\,z_{175} \end{aligned}$$

$$\begin{aligned} R_{148} = {} & 1 + 2\,z_1 + z_1\,z_6 + z_1\,z_7 + z_1\,z_{188} \\ & + z_6 + z_6\,z_{171} + z_7 + z_7\,z_{171} - z_{157}\,z_{176} \\ & + z_{168} + 2\,z_{171} + z_{171}\,z_{188} - z_{172} + z_{188} \\ & - z_{279} - z_{280} \end{aligned}$$

$$\begin{aligned} R_{149} = {} & -z_0\,z_{123} - z_0\,z_{124} - z_4\,z_{123} - z_4\,z_{124} - 12\,z_{123} \\ & - z_{123}\,z_{180} - 12\,z_{124} - z_{124}\,z_{180} + z_{181} \end{aligned}$$

$$\begin{aligned} R_{150} = {} & 143 + 24\,z_0 + z_0^2 + 2\,z_0\,z_4 + 2\,z_0\,z_{180} \\ & + 24\,z_4 + z_4^2 + 2\,z_4\,z_{180} + z_{177}^2 + 24\,z_{180} \\ & + z_{180}^2 - z_{181}^2 \end{aligned}$$

$$R_{151} = -z_{134}\,z_{179} - z_{179} + z_{182}$$

$$R_{152} = -1 + z_{178}^2 + z_{179}^2 - z_{182}^2$$

$$R_{153} = -z_{129}\,z_{183} - z_{177} + z_{178}$$

$$\begin{aligned} R_{154} = {} & -12 - z_0 - 4\,z_0\,z_{184} - z_4 - 4\,z_4\,z_{184} \\ & + z_{179} - 4\,z_{180}\,z_{184} - 48\,z_{184} \end{aligned}$$

$$\begin{aligned} R_{155} = {} & 35 + 3\,z_0 + z_0\,z_6 + z_0\,z_7 + z_0\,z_{188} \\ & + 3\,z_4 + z_4\,z_6 + z_4\,z_7 + z_4\,z_{188} + 11\,z_6 \\ & + z_6\,z_{180} - z_6\,z_{186} - z_6\,z_{187} + 11\,z_7 + z_7\,z_{180} \\ & - z_7\,z_{186} - z_7\,z_{187} + z_{177} + 3\,z_{180} + z_{180}\,z_{188} \\ & - z_{181} - z_{185} - z_{185}\,z_{211} - z_{185}\,z_{212} - 2\,z_{186} \\ & - z_{187} + 12\,z_{188} \end{aligned}$$

$$R_{156} = -12 - z_0 - z_2^2 + z_{190} - z_{193}$$

$$R_{157} = -z_{123}\,z_{190} - z_{124}\,z_{190} + z_{194}$$

$$R_{158} = -1 + z_{189}^2 + z_{190}^2 - z_{194}^2$$

$$R_{159} = -z_{134}\,z_{192} - z_{192} + z_{195}$$

$$R_{160} = -1 + z_{191}^2 + z_{192}^2 - z_{195}^2$$

$$R_{161} = -z_{129}\,z_{196} - z_{189} + z_{191}$$

$$R_{162} = -12 - z_0 - z_2^2 - 4\,z_{190}\,z_{197} + z_{192}$$

$$R_{163} = 1 + z_6 + z_7 - z_{21} + z_{188} - z_{201}$$

$$\begin{aligned} R_{164} = {} & z_6\,z_{190} + z_7\,z_{190} - z_{21} - z_{21}\,z_{186} - z_{21}\,z_{187} \\ & + z_{188}\,z_{190} + z_{189} + 3\,z_{190} - z_{194} - z_{198} \\ & - z_{198}\,z_{211} - z_{198}\,z_{212} - z_{199} \end{aligned}$$

$$R_{165} = z_{186} + z_{187} - z_{199} - z_{200}$$

$$\begin{aligned} R_{166} = {} & -z_{12}\,z_{123} - z_{12}\,z_{124} - z_{123} - z_{123}\,z_{205} - z_{124} \\ & - z_{124}\,z_{205} + z_{206} \end{aligned}$$

$$\begin{aligned} R_{167} = {} & 2\,z_{12} + z_{12}^2 + 2\,z_{12}\,z_{205} + z_{202}^2 + 2\,z_{205} \\ & + z_{205}^2 - z_{206}^2 \end{aligned}$$

$$R_{168} = -z_{134}\,z_{204} - z_{204} + z_{207}$$

$$R_{169} = -1 + z_{203}^2 + z_{204}^2 - z_{207}^2$$

$$R_{170} = -z_{129}\,z_{208} - z_{202} + z_{203}$$

$$\begin{aligned} R_{171} = {} & -1 - z_{12} - 4\,z_{12}\,z_{209} + z_{204} - 4\,z_{205}\,z_{209} \\ & - 4\,z_{209} \end{aligned}$$

$$R_{172} = -z_{22} - z_{22}\,z_{211} - z_{22}\,z_{212} + z_{213}$$

$$\begin{aligned} R_{173} = {} & 3 + z_6 + z_6\,z_{12} + z_6\,z_{205} + z_7 \\ & + z_7\,z_{12} + z_7\,z_{205} + 3\,z_{12} + z_{12}\,z_{188} + z_{188} \\ & + z_{188}\,z_{205} + z_{202} + 3\,z_{205} - z_{206} - z_{210}\,z_{213} \\ & - z_{211} \end{aligned}$$

$$\begin{aligned} R_{174} = {} & -z_{13}\,z_{123} - z_{13}\,z_{124} - z_{123} - z_{123}\,z_{217} - z_{124} \\ & - z_{124}\,z_{217} + z_{218} \end{aligned}$$

$$\begin{aligned} R_{175} = {} & 2\,z_{13} + z_{13}^2 + 2\,z_{13}\,z_{217} + z_{214}^2 + 2\,z_{217} \\ & + z_{217}^2 - z_{218}^2 \end{aligned}$$

$$R_{176} = -z_{134}\,z_{216} - z_{216} + z_{219}$$

$$R_{177} = -1 + z_{215}^2 + z_{216}^2 - z_{219}^2$$

$$R_{178} = -z_{129}\,z_{220} - z_{214} + z_{215}$$

$$R_{179} = -1 - z_{13} - 4\,z_{13}\,z_{221} + z_{216} - 4\,z_{217}\,z_{221} - 4\,z_{221}$$

$$R_{180} = 3 + z_6 + z_6\,z_{13} + z_6\,z_{217} + z_7 + z_7\,z_{13} + z_7\,z_{217} + 3\,z_{13} + z_{13}\,z_{188} + z_{188} + z_{188}\,z_{217} - z_{213}\,z_{222} + z_{214} + 3\,z_{217} - z_{218} - z_{223}$$

$$R_{181} = z_{211} + z_{212} - z_{223} - z_{224}$$

$$R_{182} = -z_{14}\,z_{123} - z_{14}\,z_{124} - z_{123} - z_{123}\,z_{228} - z_{124} - z_{124}\,z_{228} + z_{229}$$

$$R_{183} = 2\,z_{14} + z_{14}^2 + 2\,z_{14}\,z_{228} + z_{225}^2 + 2\,z_{228} + z_{228}^2 - z_{229}^2$$

$$R_{184} = -z_{134}\,z_{227} - z_{227} + z_{230}$$

$$R_{185} = -1 + z_{226}^2 + z_{227}^2 - z_{230}^2$$

$$R_{186} = -z_{129}\,z_{231} - z_{225} + z_{226}$$

$$R_{187} = -1 - z_{14} - 4\,z_{14}\,z_{232} + z_{227} - 4\,z_{228}\,z_{232} - 4\,z_{232}$$

$$R_{188} = 3 + z_6 + z_6\,z_{14} + z_6\,z_{228} + z_7 + z_7\,z_{14} + z_7\,z_{228} + 3\,z_{14} + z_{14}\,z_{188} + z_{188} + z_{188}\,z_{228} - z_{213}\,z_{233} + z_{225} + 3\,z_{228} - z_{229} - z_{234}$$

$$R_{189} = z_{211} + z_{212} - z_{234} - z_{235}$$

$$R_{190} = -z_{15}\,z_{123} - z_{15}\,z_{124} - z_{123} - z_{123}\,z_{239} - z_{124} - z_{124}\,z_{239} + z_{240}$$

$$R_{191} = 2\,z_{15} + z_{15}^2 + 2\,z_{15}\,z_{239} + z_{236}^2 + 2\,z_{239} + z_{239}^2 - z_{240}^2$$

$$R_{192} = -z_{134}\,z_{238} - z_{238} + z_{241}$$

$$R_{193} = -1 + z_{237}^2 + z_{238}^2 - z_{241}^2$$

$$R_{194} = -z_{129}\,z_{242} - z_{236} + z_{237}$$

$$R_{195} = -1 - z_{15} - 4\,z_{15}\,z_{243} + z_{238} - 4\,z_{239}\,z_{243} - 4\,z_{243}$$

$$R_{196} = 3 + z_6 + z_6\,z_{15} + z_6\,z_{239} + z_7 + z_7\,z_{15} + z_7\,z_{239} + 3\,z_{15} + z_{15}\,z_{188} + z_{188} + z_{188}\,z_{239} - z_{213}\,z_{244} + z_{236} + 3\,z_{239} - z_{240} - z_{245}$$

$$R_{197} = z_{211} + z_{212} - z_{245} - z_{246}$$

$$R_{198} = -z_{16}\,z_{123} - z_{16}\,z_{124} - z_{123} - z_{123}\,z_{250} - z_{124} - z_{124}\,z_{250} + z_{251}$$

$$R_{199} = 2\,z_{16} + z_{16}^2 + 2\,z_{16}\,z_{250} + z_{247}^2 + 2\,z_{250} + z_{250}^2 - z_{251}^2$$

$$R_{200} = -z_{134}\,z_{249} - z_{249} + z_{252}$$

$$R_{201} = -1 + z_{248}^2 + z_{249}^2 - z_{252}^2$$

$$R_{202} = -z_{129}\,z_{253} - z_{247} + z_{248}$$

$$R_{203} = -1 - z_{16} - 4\,z_{16}\,z_{254} + z_{249} - 4\,z_{250}\,z_{254} - 4\,z_{254}$$

$$R_{204} = -z_{22} - z_{22}\,z_{256} - z_{22}\,z_{257} + z_{258}$$

$$R_{205} = 3 + z_6 + z_6\,z_{16} + z_6\,z_{250} + z_7 + z_7\,z_{16} + z_7\,z_{250} + 3\,z_{16} + z_{16}\,z_{188} + z_{188} + z_{188}\,z_{250} + z_{247} + 3\,z_{250} - z_{251} - z_{255}\,z_{258} - z_{256}$$

$$R_{206} = -z_{17}\,z_{123} - z_{17}\,z_{124} - z_{123} - z_{123}\,z_{262} - z_{124} - z_{124}\,z_{262} + z_{263}$$

$$R_{207} = 2\,z_{17} + z_{17}^2 + 2\,z_{17}\,z_{262} + z_{259}^2 + 2\,z_{262} + z_{262}^2 - z_{263}^2$$

$$R_{208} = -z_{134}\,z_{261} - z_{261} + z_{264}$$

$$R_{209} = -1 + z_{260}^2 + z_{261}^2 - z_{264}^2$$

$$R_{210} = -z_{129}\,z_{265} - z_{259} + z_{260}$$

$$R_{211} = -1 - z_{17} - 4\,z_{17}\,z_{266} + z_{261} - 4\,z_{262}\,z_{266} - 4\,z_{266}$$

$$R_{212} = 3 + z_6 + z_6\,z_{17} + z_6\,z_{262} + z_7 + z_7\,z_{17} + z_7\,z_{262} + 3\,z_{17} + z_{17}\,z_{188} + z_{188} + z_{188}\,z_{262} - z_{258}\,z_{267} + z_{259} + 3\,z_{262} - z_{263} - z_{268}$$

$$R_{213} = z_{256} + z_{257} - z_{268} - z_{269}$$

$$\begin{aligned} R_{214} = & - z_{18}\, z_{123} - z_{18}\, z_{124} - z_{123} - z_{123}\, z_{273} - z_{124} \\ & - z_{124}\, z_{273} + z_{274} \end{aligned}$$

$$\begin{aligned} R_{215} = & \, 2\, z_{18} + z_{18}^2 + 2\, z_{18}\, z_{273} + z_{270}^2 + 2\, z_{273} \\ & + z_{273}^2 - z_{274}^2 \end{aligned}$$

$$R_{216} = - z_{134}\, z_{272} - z_{272} + z_{275}$$

$$R_{217} = - 1 + z_{271}^2 + z_{272}^2 - z_{275}^2$$

$$R_{218} = - z_{129}\, z_{276} - z_{270} + z_{271}$$

$$\begin{aligned} R_{219} = & - 1 - z_{18} - 4\, z_{18}\, z_{277} + z_{272} - 4\, z_{273}\, z_{277} \\ & - 4\, z_{277} \end{aligned}$$

$$\begin{aligned} R_{220} = & \, 3 + z_6 + z_6\, z_{18} + z_6\, z_{273} + z_7 \\ & + z_7\, z_{18} + z_7\, z_{273} + 3\, z_{18} + z_{18}\, z_{188} - z_{122}\, z_{278} \\ & + z_{188} + z_{188}\, z_{273} + z_{270} + 3\, z_{273} - z_{274} \\ & - z_{279} \end{aligned}$$

$$\begin{aligned} R_{221} = & - z_{19}\, z_{123} - z_{19}\, z_{124} - z_{123} - z_{123}\, z_{284} - z_{124} \\ & - z_{124}\, z_{284} + z_{285} \end{aligned}$$

$$\begin{aligned} R_{222} = & \, 2\, z_{19} + z_{19}^2 + 2\, z_{19}\, z_{284} + z_{281}^2 + 2\, z_{284} \\ & + z_{284}^2 - z_{285}^2 \end{aligned}$$

$$R_{223} = - z_{134}\, z_{283} - z_{283} + z_{286}$$

$$R_{224} = - 1 + z_{282}^2 + z_{283}^2 - z_{286}^2$$

$$R_{225} = - z_{129}\, z_{287} - z_{281} + z_{282}$$

$$\begin{aligned} R_{226} = & - 1 - z_{19} - 4\, z_{19}\, z_{288} + z_{283} - 4\, z_{284}\, z_{288} \\ & - 4\, z_{288} \end{aligned}$$

$$\begin{aligned} R_{227} = & \, 3 + z_6 + z_6\, z_{19} + z_6\, z_{284} + z_7 \\ & + z_7\, z_{19} + z_7\, z_{284} + 3\, z_{19} + z_{19}\, z_{188} - z_{122}\, z_{289} \\ & + z_{188} + z_{188}\, z_{284} + z_{281} + 3\, z_{284} - z_{285} \\ & - z_{290} \end{aligned}$$

$$R_{228} = z_{279} + z_{280} - z_{290} - z_{291}$$

$$\begin{aligned} R_{229} = & - z_{20}\, z_{123} - z_{20}\, z_{124} - z_{123} - z_{123}\, z_{295} - z_{124} \\ & - z_{124}\, z_{295} + z_{296} \end{aligned}$$

$$R_{230} = 2\,z_{20} + z_{20}^2 + 2\,z_{20}\,z_{295} + z_{292}^2 + 2\,z_{295} + z_{295}^2 - z_{296}^2$$

$$R_{231} = -z_{134}\,z_{294} - z_{294} + z_{297}$$

$$R_{232} = -1 + z_{293}^2 + z_{294}^2 - z_{297}^2$$

$$R_{233} = -z_{129}\,z_{298} - z_{292} + z_{293}$$

$$R_{234} = -1 - z_{20} - 4\,z_{20}\,z_{299} + z_{294} - 4\,z_{295}\,z_{299} - 4\,z_{299}$$

$$R_{235} = -z_{22} - z_{22}\,z_{186} - z_{22}\,z_{187} + z_{303}$$

$$R_{236} = 3 + z_6 + z_6\,z_{20} + z_6\,z_{295} + z_7 + z_7\,z_{20} + z_7\,z_{295} + 3\,z_{20} + z_{20}\,z_{188} + z_{188} + z_{188}\,z_{295} + z_{292} + 3\,z_{295} - z_{296} - z_{300}\,z_{303} - z_{301}$$

$$R_{237} = z_{186} + z_{187} - z_{301} - z_{302}$$

$$R_{238} = -z_{12}\,z_{123} - z_{12}\,z_{124} - 13\,z_{13}\,z_{123} - 13\,z_{13}\,z_{124} - 4\,z_{24}\,z_{123} - 4\,z_{24}\,z_{124} - 2\,z_{27}\,z_{123} - 2\,z_{27}\,z_{124} - z_{31}\,z_{123} - z_{31}\,z_{124} - z_{33}\,z_{123} - z_{33}\,z_{124} - z_{35}\,z_{123} - z_{35}\,z_{124} - z_{37}\,z_{123} - z_{37}\,z_{124} - 2\,z_{41}\,z_{123} - 2\,z_{41}\,z_{124} - 4\,z_{43}\,z_{123} - 4\,z_{43}\,z_{124} - 2\,z_{44}\,z_{123} - 2\,z_{44}\,z_{124} - z_{47}\,z_{123} - z_{47}\,z_{124} - 2\,z_{69}\,z_{123} - 2\,z_{69}\,z_{124} - 25\,z_{123} - z_{123}\,z_{307} - 25\,z_{124} - z_{124}\,z_{307} + z_{308}$$

$$
\begin{aligned}
R_{239} = {} & 624 + 50\,z_{12} + z_{12}^2 + 26\,z_{12}\,z_{13} + 8\,z_{12}\,z_{24} \\
& + 4\,z_{12}\,z_{27} + 2\,z_{12}\,z_{31} + 2\,z_{12}\,z_{33} + 2\,z_{12}\,z_{35} + 2\,z_{12}\,z_{37} \\
& + 4\,z_{12}\,z_{41} + 8\,z_{12}\,z_{43} + 4\,z_{12}\,z_{44} + 2\,z_{12}\,z_{47} + 4\,z_{12}\,z_{69} \\
& + 2\,z_{12}\,z_{307} + 650\,z_{13} + 169\,z_{13}^2 + 104\,z_{13}\,z_{24} + 52\,z_{13}\,z_{27} \\
& + 26\,z_{13}\,z_{31} + 26\,z_{13}\,z_{33} + 26\,z_{13}\,z_{35} + 26\,z_{13}\,z_{37} + 52\,z_{13}\,z_{41} \\
& + 104\,z_{13}\,z_{43} + 52\,z_{13}\,z_{44} + 26\,z_{13}\,z_{47} + 52\,z_{13}\,z_{69} + 26\,z_{13}\,z_{307} \\
& + 200\,z_{24} + 16\,z_{24}^2 + 16\,z_{24}\,z_{27} + 8\,z_{24}\,z_{31} + 8\,z_{24}\,z_{33} \\
& + 8\,z_{24}\,z_{35} + 8\,z_{24}\,z_{37} + 16\,z_{24}\,z_{41} + 32\,z_{24}\,z_{43} + 16\,z_{24}\,z_{44} \\
& + 8\,z_{24}\,z_{47} + 16\,z_{24}\,z_{69} + 8\,z_{24}\,z_{307} + 100\,z_{27} + 4\,z_{27}^2 \\
& + 4\,z_{27}\,z_{31} + 4\,z_{27}\,z_{33} + 4\,z_{27}\,z_{35} + 4\,z_{27}\,z_{37} + 8\,z_{27}\,z_{41} \\
& + 16\,z_{27}\,z_{43} + 8\,z_{27}\,z_{44} + 4\,z_{27}\,z_{47} + 8\,z_{27}\,z_{69} + 4\,z_{27}\,z_{307} \\
& + 50\,z_{31} + z_{31}^2 + 2\,z_{31}\,z_{33} + 2\,z_{31}\,z_{35} + 2\,z_{31}\,z_{37} \\
& + 4\,z_{31}\,z_{41} + 8\,z_{31}\,z_{43} + 4\,z_{31}\,z_{44} + 2\,z_{31}\,z_{47} + 4\,z_{31}\,z_{69} \\
& + 2\,z_{31}\,z_{307} + 50\,z_{33} + z_{33}^2 + 2\,z_{33}\,z_{35} + 2\,z_{33}\,z_{37} \\
& + 4\,z_{33}\,z_{41} + 8\,z_{33}\,z_{43} + 4\,z_{33}\,z_{44} + 2\,z_{33}\,z_{47} + 4\,z_{33}\,z_{69} \\
& + 2\,z_{33}\,z_{307} + 50\,z_{35} + z_{35}^2 + 2\,z_{35}\,z_{37} + 4\,z_{35}\,z_{41} \\
& + 8\,z_{35}\,z_{43} + 4\,z_{35}\,z_{44} + 2\,z_{35}\,z_{47} + 4\,z_{35}\,z_{69} + 2\,z_{35}\,z_{307} \\
& + 50\,z_{37} + z_{37}^2 + 4\,z_{37}\,z_{41} + 8\,z_{37}\,z_{43} + 4\,z_{37}\,z_{44} \\
& + 2\,z_{37}\,z_{47} + 4\,z_{37}\,z_{69} + 2\,z_{37}\,z_{307} + 100\,z_{41} + 4\,z_{41}^2 \\
& + 16\,z_{41}\,z_{43} + 8\,z_{41}\,z_{44} + 4\,z_{41}\,z_{47} + 8\,z_{41}\,z_{69} + 4\,z_{41}\,z_{307} \\
& + 200\,z_{43} + 16\,z_{43}^2 + 16\,z_{43}\,z_{44} + 8\,z_{43}\,z_{47} + 16\,z_{43}\,z_{69} \\
& + 8\,z_{43}\,z_{307} + 100\,z_{44} + 4\,z_{44}^2 + 4\,z_{44}\,z_{47} + 8\,z_{44}\,z_{69} \\
& + 4\,z_{44}\,z_{307} + 50\,z_{47} + z_{47}^2 + 4\,z_{47}\,z_{69} + 2\,z_{47}\,z_{307} \\
& + 100\,z_{69} + 4\,z_{69}^2 + 4\,z_{69}\,z_{307} + z_{304}^2 + 50\,z_{307} \\
& + z_{307}^2 - z_{308}^2
\end{aligned}
$$

$$
R_{240} = -z_{134}\,z_{306} - z_{306} + z_{309}
$$

$$
R_{241} = -1 + z_{305}^2 + z_{306}^2 - z_{309}^2
$$

$$
R_{242} = -z_{129}\,z_{310} - z_{304} + z_{305}
$$

$$
\begin{aligned}
R_{243} = {} & -25 - z_{12} - 4\,z_{12}\,z_{311} - 13\,z_{13} - 52\,z_{13}\,z_{311} \\
& - 4\,z_{24} - 16\,z_{24}\,z_{311} - 2\,z_{27} - 8\,z_{27}\,z_{311} - z_{31} \\
& - 4\,z_{31}\,z_{311} - z_{33} - 4\,z_{33}\,z_{311} - z_{35} - 4\,z_{35}\,z_{311} \\
& - z_{37} - 4\,z_{37}\,z_{311} - 2\,z_{41} - 8\,z_{41}\,z_{311} - 4\,z_{43} \\
& - 16\,z_{43}\,z_{311} - 2\,z_{44} - 8\,z_{44}\,z_{311} - z_{47} - 4\,z_{47}\,z_{311} \\
& - 2\,z_{69} - 8\,z_{69}\,z_{311} + z_{306} - 4\,z_{307}\,z_{311} - 100\,z_{311}
\end{aligned}
$$

$$
\begin{aligned}
R_{244} = 75 &+ 25\,z_6 + z_6\,z_{12} + 13\,z_6\,z_{13} + 4\,z_6\,z_{24} \\
&+ 2\,z_6\,z_{27} + z_6\,z_{31} + z_6\,z_{33} + z_6\,z_{35} + z_6\,z_{37} \\
&+ 2\,z_6\,z_{41} + 4\,z_6\,z_{43} + 2\,z_6\,z_{44} + z_6\,z_{47} + 2\,z_6\,z_{69} \\
&+ z_6\,z_{307} + 25\,z_7 + z_7\,z_{12} + 13\,z_7\,z_{13} + 4\,z_7\,z_{24} \\
&+ 2\,z_7\,z_{27} + z_7\,z_{31} + z_7\,z_{33} + z_7\,z_{35} + z_7\,z_{37} \\
&+ 2\,z_7\,z_{41} + 4\,z_7\,z_{43} + 2\,z_7\,z_{44} + z_7\,z_{47} + 2\,z_7\,z_{69} \\
&+ z_7\,z_{307} + 3\,z_{12} + z_{12}\,z_{188} + 39\,z_{13} + 13\,z_{13}\,z_{188} \\
&+ 12\,z_{24} + 4\,z_{24}\,z_{188} + 6\,z_{27} + 2\,z_{27}\,z_{188} + 3\,z_{31} \\
&+ z_{31}\,z_{188} + 3\,z_{33} + z_{33}\,z_{188} + 3\,z_{35} + z_{35}\,z_{188} \\
&+ 3\,z_{37} + z_{37}\,z_{188} + 6\,z_{41} + 2\,z_{41}\,z_{188} + 12\,z_{43} \\
&+ 4\,z_{43}\,z_{188} + 6\,z_{44} + 2\,z_{44}\,z_{188} + 3\,z_{47} + z_{47}\,z_{188} \\
&+ 6\,z_{69} + 2\,z_{69}\,z_{188} + 25\,z_{188} + z_{188}\,z_{307} - z_{213}\,z_{312} \\
&+ z_{304} + 3\,z_{307} - z_{308} - z_{313}
\end{aligned}
$$

$$
R_{245} = z_{211} + z_{212} - z_{313} - z_{314}
$$

$$
\begin{aligned}
R_{246} = &- 360 - 54\,z_0 - 2\,z_0^2 - 2\,z_0\,z_1 - 4\,z_0\,z_2 \\
&- 2\,z_0\,z_9 - 2\,z_0\,z_{138} - z_0\,z_{148} - 24\,z_1 - 2\,z_1\,z_{138} \\
&- 48\,z_2 - 4\,z_2\,z_{138} - 24\,z_9 - 2\,z_9\,z_{138} - 30\,z_{138} \\
&- z_{138}\,z_{148} - 12\,z_{148} + z_{315}
\end{aligned}
$$

$$
R_{247} = - z_2\,z_{315} - z_9\,z_{315} - z_{162}\,z_{315} - 2\,z_{315} + z_{316}
$$

$$
R_{248} = - z_1\,z_{316} - z_{171}\,z_{316} - z_{316} + z_{317}
$$

$$
R_{249} = - z_0\,z_{317} - z_4\,z_{317} - z_{180}\,z_{317} - 12\,z_{317} + z_{318}
$$

$$
R_{250} = - z_{190}\,z_{318} + z_{319}
$$

$$
R_{251} = - z_{12}\,z_{319} - z_{205}\,z_{319} - z_{319} + z_{320}
$$

$$
R_{252} = - z_{13}\,z_{320} - z_{217}\,z_{320} - z_{320} + z_{321}
$$

$$
R_{253} = - z_{14}\,z_{321} - z_{228}\,z_{321} - z_{321} + z_{322}
$$

$$
R_{254} = - z_{15}\,z_{322} - z_{239}\,z_{322} - z_{322} + z_{323}
$$

$$
R_{255} = - z_{16}\,z_{323} - z_{250}\,z_{323} - z_{323} + z_{324}
$$

$$
R_{256} = - z_{17}\,z_{324} - z_{262}\,z_{324} - z_{324} + z_{325}
$$

$$
R_{257} = - z_{18}\,z_{325} - z_{273}\,z_{325} - z_{325} + z_{326}
$$

$$
R_{258} = - z_{19}\,z_{326} - z_{284}\,z_{326} - z_{326} + z_{327}
$$

$$
R_{259} = - z_{20}\,z_{327} - z_{295}\,z_{327} - z_{327} + z_{328}
$$

$$
\begin{aligned}
R_{260} = &- z_{12}\,z_{328} - 13\,z_{13}\,z_{328} - 4\,z_{24}\,z_{328} - 2\,z_{27}\,z_{328} - z_{31}\,z_{328} \\
&- z_{33}\,z_{328} - z_{35}\,z_{328} - z_{37}\,z_{328} - 2\,z_{41}\,z_{328} - 4\,z_{43}\,z_{328} \\
&- 2\,z_{44}\,z_{328} - z_{47}\,z_{328} - 2\,z_{69}\,z_{328} - z_{307}\,z_{328} - 25\,z_{328} \\
&+ z_{329}
\end{aligned}
$$

$$
R_{261} = - z_{329}^2 + z_{330}
$$

$$
R_{262} = z_{130} - z_{330} - z_{330}\,z_{331}
$$